

HW04: Fairness and Bias Detection with AIF360

Course: CS 516 — Responsible Data Science and Algorithmic Fairness

Due Date: November 14, 2025 – 11:59 PM

Objective:

In this homework, you will explore **bias detection and mitigation** using IBM's **AI Fairness 360 (AIF360)** toolkit.

You'll analyze the **German Credit dataset** to see how bias affects predictions and apply the **Reweighting algorithm** to make outcomes fairer.

What to Submit:

Download the provided HW4.ipynb notebook from Gradescope.

👉 [HW-04:Fairness and Bias Detection with AIF360.ipynb](#)

1. **Complete all TODO cells** following the step-by-step instructions.
2. **Answer all short reflection questions (Q1–Q4)** clearly and concisely.
3. **Run all cells** to verify your code and outputs are correct.
4. **Export your notebook to PDF:**
 - In Colab → *File* → *Download* → *Download PDF*
 - OR *File* → *Print* → *Save as PDF*
5. **Submit only the PDF** on Gradescope under **"HW4 – Fairness & Bias Detection."**

Grading Rubric:

Step	Task Description	Points
Step 1	Import all required libraries correctly	10
Step 2	Load dataset, set protected attribute, split into train/test	15
Step 3	Compute fairness metrics (Mean Difference, Pearson Correlation)	20
Q1–Q2	Conceptual answers about fairness interpretation	10
Step 4	Implement Reweighting algorithm correctly	15
Q3	Explain the Reweighting concept and its purpose	10
Step 5	Compute fairness metric after mitigation	10
Q4	Reflection on bias mitigation effectiveness	10
Total	—	100

Best

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